

STR – Fruit Splitter Information – Luís Romba

UI and System functionalities



Figure 1: System and UI Interactions

The UI gives various functionalities to help the students explore and test the system:

- Apple Button – Deploy an apple on the start station;
- Pear Button – Deploy a pear on the start station;
- Lemon Button – Deploy a lemon on the start station;
- Clean Button – Clean all fruits;
- Force Slide – It's possible to change the speed/force of the actuators movement. (Carefully: Sometimes if the force is too high, the system may bug some sensors depending on the student pc, in that case the simulation web page needs to be restarted or even in some cases the .py running on the console may need to be restarted)
- Menu Button – Returns to the Menu where it can select other systems given as examples;
- Reset – It resets all actuators positions to the start as if the simulation has been restarted;
- Free Cam Button – It lets freely move the camera around the system while activated (keyboard buttons a, w, s and d to move the camera and right click the mouse to rotate the camera);
- View Buttons – Various views that can help see the system working from different angles. (Curiosity: If one of the views buttons and free cam button are activated at the same time, camera is moved, and that view button turn off after then the angle of that view will change to that last moment when the button was turn off for the rest of the simulation);

- Reset Views Button – If the angles of the views have been changed then it will reset all views with the angles by default at the start of the simulation;
- Sensors Button – Shows all pins associated with the sensors and their actual value. It's possible to manually change their value by clicking on the value label to test the system.
- Actuators Button – Shows all pins associated with the actuator's movement and their actual value. It's possible to change manually their value label by clicking on the value to test the system;
- Clicking Sensors and UI show information – By clicking on the sensors, they will turn red meaning the information at the top represents that sensor. If the sensor is green, it means that the value of the associated pin is 1 (high) and by clicking on it to show the information, in this case, it will turn pink instead of red.
- Clicking Actuators Motor and UI show information – Depending on the pins associated with the actuators movement that are high, the wall in the stations will start moving and depending on the direction, the actuator motor will start rotating clockwise (backward) or anti clockwise (forward). By clicking on the motors, the information at the top will say which pins make the cylinders move forward and backward (Conveyor just moves forward);

Sensors, Actuators and Pins

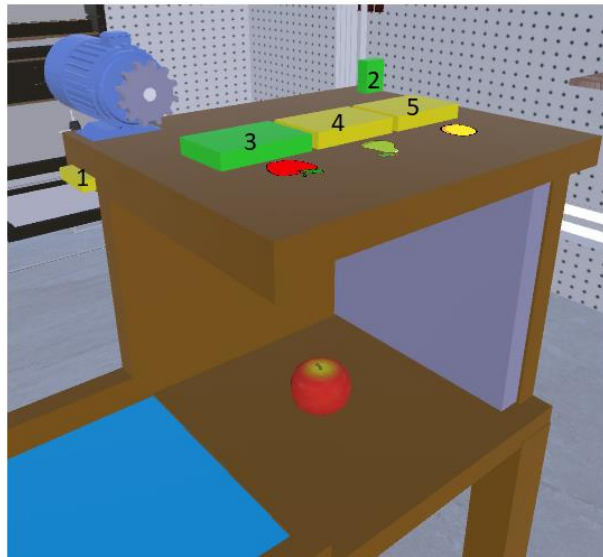


Figure 2: Start Station/ Station 1 – Sensors and associated Pins

- Start Station or Station 1 is where the system starts by putting fruits (apples, pears or lemons) manually with the given UI buttons or keyboard associated with the pins 20 for apples, 21 for pears and 22 for lemons (in case of keyboard use, it needs to be made in the python file where the students develop).
- Sensor with pin 1 will be high when the cylinder is at the work position and pin 2 will be high when the cylinder is at the rest position.
- Sensor 3, 4 and 5 will be high when it detects apple, pear and lemon, respectively at the zone shown in figure 2. (Example – Pin 3 is high because it's an apple inside the station).

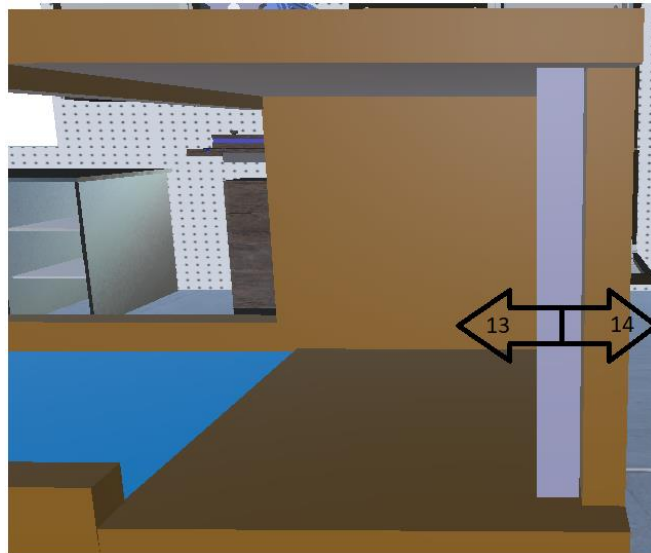


Figure 3: Start Station/Station 1 – Cylinder movement

- In the case of the Start Station/Station 1, if the value of the pin 13 is high then it will activate the actuator to move the cylinder forward and if the pin 14 is high then it will go backwards.

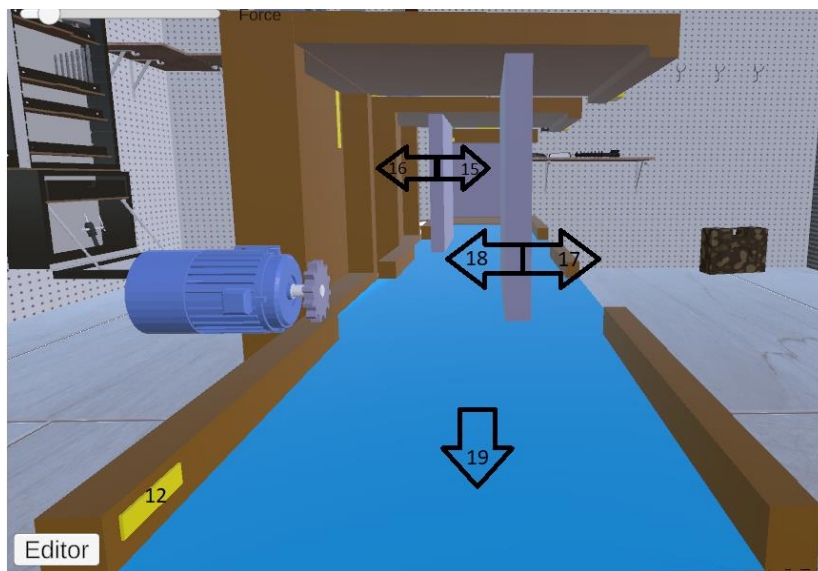


Figure 4: Station 2, Station 3 and conveyor – Cylinders movement and Sensor/Pin 12.

- In the case of Station 2, if the value of the pin 15 is high then it will activate the actuator to move the cylinder forward and if the pin 16 is high then it will go backwards.
- In the case of Station 3, if the value of the pin 17 is high then it will activate the actuator to move the cylinder forward and if the pin 18 is high then it will go backwards.
- The fruits at the conveyor will move forward when the actuator is activated by putting the value high on the pin 19.
- Pin 12 is high when the sensor detects any fruit in its respective detect zone.

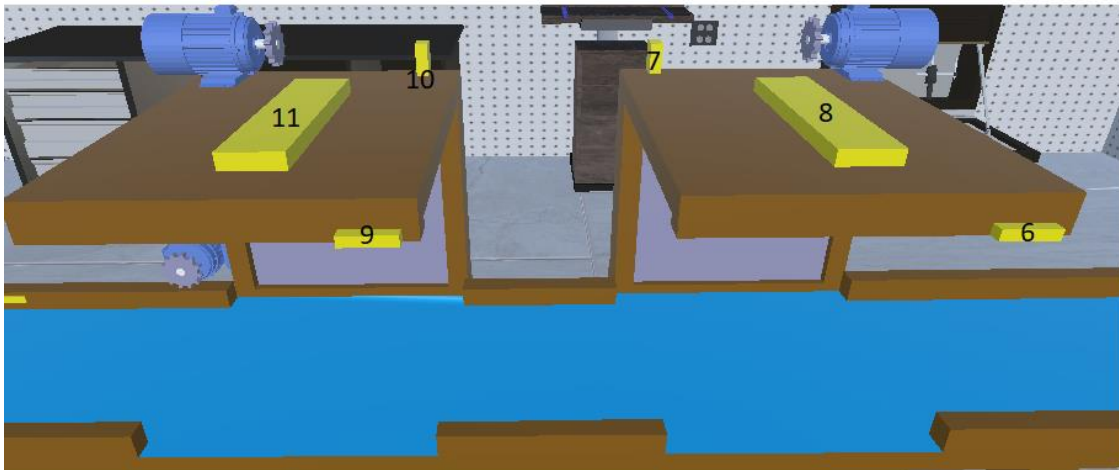


Figure 5: Station 2 and 3 – Sensors and associated pins.

- Sensors with pins 6 and 9 will be high when their respective cylinder is at the work position and pins 7 and 10 will be high when their respective cylinder is at the rest position.
- Sensors with pins 8 and 11 detect any fruit on the conveyor at their respective stations and detect zone.
- When pins 23 (box at station 2), 24 (box at station 3) and 25 (box at the end of the conveyor) are high, it makes the respective boxes and fruits in them disappear.

Resume of Pins

```
GPIO.setup(1, GPIO.INPUT) # cylinder 1, WORK position
GPIO.setup(2, GPIO.INPUT) # cylinder 1, REST position
GPIO.setup(3, GPIO.INPUT) # Detects Apple at station 1
GPIO.setup(4, GPIO.INPUT) # Detects Pear at station 1
GPIO.setup(5, GPIO.INPUT) # Detects Lemon at station 1

GPIO.setup(6, GPIO.INPUT) # cylinder 2, WORK position
GPIO.setup(7, GPIO.INPUT) # cylinder 2, REST position
GPIO.setup(8, GPIO.INPUT) # detects Fruit at station 2

GPIO.setup(9, GPIO.INPUT) # cylinder 3, WORK position
GPIO.setup(10, GPIO.INPUT) # cylinder 3, REST position
GPIO.setup(11, GPIO.INPUT) # detects fruit at station 3

GPIO.setup(12, GPIO.INPUT) # detects fruit at station 4

GPIO.setup(13, GPIO.OUTPUT, GPIO.LOW) # cylinder 1, move forward
GPIO.setup(14, GPIO.OUTPUT, GPIO.LOW) # cylinder 1, move backward
GPIO.setup(15, GPIO.OUTPUT, GPIO.LOW) # cylinder 2, move forward
GPIO.setup(16, GPIO.OUTPUT, GPIO.LOW) # cylinder 2, move backward
GPIO.setup(17, GPIO.OUTPUT, GPIO.LOW) # cylinder 3, move forward
GPIO.setup(18, GPIO.OUTPUT, GPIO.LOW) # cylinder 3, move backward

GPIO.setup(19, GPIO.OUTPUT, GPIO.LOW) # conveyor moves forward

GPIO.setup(20, GPIO.OUTPUT, GPIO.LOW) # spawn apple
GPIO.setup(21, GPIO.OUTPUT, GPIO.LOW) # spawn pear
GPIO.setup(22, GPIO.OUTPUT, GPIO.LOW) # spawn lemon

GPIO.setup(23, GPIO.OUTPUT, GPIO.LOW) # Del and Spawn Apple Box
GPIO.setup(24, GPIO.OUTPUT, GPIO.LOW) # Del and Spawn Pear Box
GPIO.setup(25, GPIO.OUTPUT, GPIO.LOW) # Del and Spawn Lemon Box
```